

EVEN

Group Expense Intelligence

Product Requirements Document · v1.0 · 2026

Product Name	EVEN
Version	v1.0 — Seed / MVP
Document Owner	Mann (Founder, Brand & Product)
Company	
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Confidentiality	Internal Only — Do Not Distribute

1. Executive Summary

Pact is a privacy-first, local-first group expense tracker and bill splitter built for tight-knit friend groups, roommates, and travel companions. Unlike incumbents such as Splitwise, Pact operates entirely on-device with zero mandatory account creation — making it structurally impossible for user financial data to be monetized or leaked.

The product is designed to scale from a personal-use utility into a defensible consumer startup by embedding AI-powered intelligence, visual debt graph analysis, and a group financial health score — features no funded competitor can adopt without dismantling their core data monetization model.

Strategic Insight

Minimalist's acquisition by HUL in 2025 proved that radical transparency creates lasting consumer trust in otherwise opaque industries. Pact applies the same philosophy to personal finance between friends: full transparency, zero surveillance, community-first architecture.

2. Problem Statement

2.1 The User Problem

Splitting expenses with friends is universally painful. Current tools either:

- Require all participants to create accounts, creating friction and drop-off
- Store sensitive financial data on remote servers, creating privacy risk
- Lack intelligent suggestions — forcing manual categorization and split math
- Provide no visual understanding of group financial dynamics
- Offer no recurring expense tracking, leading to forgotten obligations

2.2 The Market Gap

Splitwise dominates mindshare but is a 2011-era product built on a 2011-era privacy model. The landscape since then has changed:

Factor	Splitwise	Pact
Privacy Model	Cloud-first, data on servers	Local-first, data on device
Account Required	Yes — all users	Yes – all users
AI Intelligence	None	Smart split + receipt AI

Factor	Splitwise	Pact
Debt Visualization	Simple list	Interactive network graph
Group Health Score	None	Built-in financial wellness score
Recurring Expenses	Basic	Full scheduling + reminders
Data Monetization Risk	High	None (no server storage)
Receipt Scanning	Paid feature	Free, AI-powered

3. Product Vision & Positioning

3.1 Vision Statement

Product Vision

Pact is the expense layer for people who trust each other — not corporations. Built local-first, designed for real friendships, and powered by AI that helps, not harvests.

3.2 Positioning

- **Primary:** Category:
 - Group expense tracker + bill splitter
- **Secondary:** Privacy-first fintech for consumer social finance
- **Primary:** Target Audience:
 - 18–28 year old friend groups, college students, roommates, travel groups
 - Privacy-conscious users in India and global Gen-Z markets
- **Core Moat:** Differentiator:
 - Local-first architecture makes privacy the structural default, not a marketing claim
 - No competitor can adopt this without rebuilding their entire data infrastructure

3.3 Competitive Moat Summary

Moat Type	Description	Reversibility for Competitors
Architectural	Local-first, no cloud storage by default	Near-impossible — requires full rebuild

Moat Type	Description	Reversibility for Competitors
AI Features	Smart split suggestions + receipt scanning	Replicable but takes 12–18 months
Visual Intelligence	Debt network graph + settlement optimizer	Medium difficulty
Community Trust	Will require account to verify users	Cannot be faked once data trust is broken
Group Health Score	Novel metric — no competitor has this	Easy to copy, hard to own retroactively

4. Feature Specification

4.1 Core Features (MVP — v1.0)

4.1.1 Group & Expense Management

The foundational split-bill and expense tracking layer — feature-parity with Splitwise plus improvements.

- Create named groups with optional emoji/avatar (no sign-up required)
- Add members by name only — no email, phone, or account needed
- Add expenses: amount, description, payer, split type, date, category
- Split modes: Equal, Exact Amount, Percentage, Shares-based
- Edit and delete expenses with full history log
- Settle debts — mark payments, log UPI/cash/bank references
- Multi-currency support with manual exchange rate input
- Export group ledger as CSV or PDF

4.1.2 Local-First Privacy Architecture

Pact's defining architectural moat. No data leaves the device unless the user explicitly initiates an export.

- All data stored in browser localStorage / IndexedDB
- No user accounts, no sign-in, no email collection
- Optional encrypted JSON export for backup (user-controlled)
- Optional QR-code-based group sync for inviting friends (P2P, no server)
- Clear data button with explicit user confirmation

Privacy Promise (Core Brand Commitment)

Pact will never store user financial data on external servers without explicit opt-in. This is not a policy — it is a technical constraint baked into the architecture. There is no server to breach.

4.1.3 Debt Graph Visualization

The most visually differentiating feature of Pact. An interactive network graph showing who owes whom, with smart settlement path optimization.

- Interactive force-directed graph of group members as nodes
- Edges represent debts — thickness = magnitude, color = direction
- Smart settlement optimizer: suggests minimum transactions to clear all debts
- Example: A owes B ₹200, B owes C ₹200 → Pact suggests A pays C directly
- Tap a node to see that member's full debt breakdown
- Animate settlement when debts are marked as paid

4.1.4 AI-Powered Smart Split Suggestions

On-device AI (via Gemini Flash API) that reads context and suggests fair splits.

- Analyze expense description to suggest category automatically
- Suggest relevant group members to include based on past patterns
- Detect unequal split scenarios (e.g., dietary restrictions, non-attendees)
- Natural language expense entry: 'Dinner at Tatva, 6 people, ₹3200' → auto-fills form
- AI chat: limited queries per day (privacy-safe, no conversation stored)

4.1.5 Receipt Scanning

Upload or capture a receipt photo — AI extracts line items, total, and suggests splits.

- Camera capture or image upload
- OCR + AI extraction of: merchant, total, line items, date
- Line-item level splitting: assign different items to different people
- Works offline for camera capture; AI processing uses Gemini Flash API (free tier)
- Receipt image stored locally, never uploaded without user action

4.1.6 Recurring Expense Tracking

Track shared subscriptions, rent, utilities, and other recurring costs.

- Create recurring expenses: weekly, monthly, custom interval
- Auto-generate expense entry on due date with local notification

- Template system: save a split configuration and reuse it
- Dashboard widget showing upcoming recurring obligations
- History view: all past instances of a recurring expense

4.1.7 Friend Group Financial Health Score

A novel, proprietary metric unique to Pact. Computed entirely on-device from group expense patterns.

- Score: 0–100, displayed as a ring gauge with trend indicator
- Factors: settlement velocity, balance symmetry, expense frequency, reciprocity ratio
- Breakdown: explains what is dragging the score down
- Historical trend: 30/60/90 day score graph
- Insights: 'Your group settles within 3 days on average — top 15%'

Why This Matters

No expense tracker has ever gamified financial health at a group level. This feature creates emotional stickiness — users will check their score the way they check their credit score. It is also uniquely defensible because the scoring model is proprietary.

4.2 Secondary Features

- Dark mode (default) + light mode toggle
- Personal expense tracker (non-group, private journal of spending)
- Expense categories with custom emoji + color
- Spending analytics: charts by category, member, time
- Activity feed: chronological log of all group actions
- Search and filter expenses by date, amount, category, member

4.3 Feature Roadmap

Phase	Timeline	Features
v1.0 — MVP	Month 1–2	Group management, expense splitting, debt graph, local storage, basic UI
v1.1 — AI Layer	Month 3	Smart split suggestions, receipt scanning, natural language input
v1.2 — Intelligence	Month 4	Health score, recurring expenses, analytics dashboard
v1.3 — Sync	Month 5	P2P QR sync, encrypted backup/restore, iOS PWA

Phase	Timeline	Features
v2.0 — Scale	Month 6+	Optional cloud sync (opt-in), web app PWA, community features

5. Recommended Tech Stack

5.1 Philosophy

The stack is chosen for three non-negotiable constraints: (1) zero-budget to start, (2) maximum development velocity for a duo vibe-coder, (3) architectural alignment with privacy-first principles. Every choice must be justifiable on all three axes.

5.2 Frontend

Layer	Technology	Reason
Framework	Next.js 14 (App Router)	File-based routing, fast, great DX, free on Vercel
Language	TypeScript	Catches bugs at build time, essential for complex state
Styling	Tailwind CSS	Fastest way to build clean UI, zero CSS files
UI Components	shadcn/ui	Unstyled, composable, Linear/Notion aesthetic out of box
Animations	Framer Motion	Debt graph animations, page transitions, micro-interactions
Icons	Lucide React	Clean, consistent, matches Linear aesthetic
Fonts	Geist (Vercel)	Modern, minimal, startup-grade — perfect for Pact's brand
State Management	Zustand	Minimal, fast, no boilerplate — better than Redux for this scale
Charts	Recharts	Clean charts for analytics, health score visualization
Graph Viz	D3.js	Debt network graph — force-directed, fully customizable
Forms	React Hook Form + Zod	Fast validation, minimal re-renders

5.3 Storage (Local-First)

Layer	Technology	Reason
Primary DB	Dexie.js (IndexedDB wrapper)	Structured local DB, async, works offline, fast queries
State Persistence	Zustand + persist middleware	Syncs Zustand state to localStorage seamlessly
Backup Format	Encrypted JSON (AES-256)	User-controlled backup via Web Crypto API — no server needed
Sync (P2P)	QR code + manual JSON import	Zero-server group invite — just scan and merge

5.4 AI & Backend

Layer	Technology	Reason
AI API	Google Gemini Flash 2.0	Free tier (generous), fast, multimodal for receipt scanning
AI Gateway	Next.js API Routes	Serverless — no dedicated backend, zero cost on Vercel
Receipt OCR	Gemini Vision (multimodal)	Single API does image understanding + text extraction
NLP Input	Gemini Flash text	Natural language → structured expense JSON
Rate Limiting	Upstash Redis (free tier)	Prevent AI API abuse, enforce daily chat limits
Auth (optional)	None in v1 / Clerk in v2	No auth needed for local-first; Clerk if cloud sync added

5.5 Infrastructure & Deployment

Layer	Technology	Reason
Hosting	Vercel (Hobby — Free)	Zero cost, edge deployment, Next.js native, instant deploys
Domain	Pact.app or getpact.in	Short, brandable, memorable
Analytics	Vercel Analytics (free)	Privacy-respecting, no cookies, GDPR-safe
Error Tracking	Sentry (free tier)	Catch bugs in production without user data exposure

Layer	Technology	Reason
CI/CD	GitHub Actions + Vercel auto-deploy	Push to main → auto-deploy, zero config
Repo	GitHub (private)	Standard, free, integrates with everything

5.6 Development Tooling

Tool	Purpose
Cursor IDE	AI-assisted vibe coding — primary development environment
ESLint + Prettier	Code quality and formatting
Husky + lint-staged	Pre-commit hooks to enforce quality
pnpm	Faster than npm, better disk efficiency
Storybook (optional)	Component development and documentation

5.7 Full Stack Summary

The Stack in One Line

Next.js 14 + TypeScript + Tailwind + shadcn/ui + Zustand + Dexie.js + Gemini Flash API + Vercel.
Estimated monthly cost at launch: ₹0.

6. UX & Design System

6.1 Design Principles

- Clarity over cleverness — every screen has one primary action
- Data density is a feature, not a bug — power users should see more
- Motion with purpose — animations communicate state, not decoration
- Privacy signals everywhere — reinforce the local-first message in UI copy

6.2 Design System

Token	Value	Usage
Primary	#0F172A (Slate 900)	Brand color, headings, high-emphasis text
Accent	#6366F1 (Indigo 500)	CTAs, active states, links, health score
Success	#10B981 (Emerald 500)	Settled debts, positive indicators
Warning	#F59E0B (Amber 500)	Overdue, reminders
Destructive	#EF4444 (Red 500)	Delete actions, debt indicators
Surface	#F8FAFC (Slate 50)	Card backgrounds, inputs
Border	#E2E8F0 (Slate 200)	Dividers, card borders
Font	Geist Sans + Geist Mono	All UI text + numbers/amounts

6.3 Key Screens

- Dashboard — Group health score ring, quick-add FAB, recent activity, upcoming recurring
- Group View — Member list with balances, debt graph toggle, add expense CTA
- Debt Graph — Full-screen force graph, settlement optimizer panel, node detail drawer
- Add Expense — Smart form with AI suggestions, receipt upload zone, split configurator
- Analytics — Category donut, member bar chart, 30/60/90 day trends
- Settings — Export data, clear data, currency, theme, AI chat quota

7. Success Metrics

7.1 North Star Metric

Weekly Active Groups — the number of distinct groups that add at least one expense per week. This is the truest signal of Pact being embedded in real social behavior.

7.2 v1.0 Launch Targets (90 days post-launch)

Metric	Target	Measurement Method
Total installs (PWA)	500+	Vercel Analytics
Weekly Active Groups	100+	Local analytics ping (privacy-safe)
Avg expenses per group	8+	Indicates real usage, not just trial
Receipt scans	200+	Gemini API call logs

Metric	Target	Measurement Method
D7 retention	>40%	Return visit to app within 7 days
Health Score viewed	>60% of active users	Click event on score component
Organic referrals	30%+ of new installs	UTM tracking on share links

8. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Data loss (no cloud backup)	Medium	High	Prominent backup reminder, auto-export prompt
Gemini API costs at scale	Low (free tier generous)	Medium	Rate limit per user, upgrade to paid if needed
P2P sync complexity	High	Medium	Ship QR import in v1.1, not v1.0
Low group adoption (network effect)	Medium	High	Solo personal tracker mode to acquire users pre-group
Competitor copying health score	Medium	Low	Own the brand narrative early — be first, be vocal
iOS PWA limitations	Medium	Medium	Test thoroughly, document known limitations

9. Appendix

9.1 Glossary

- Local-first: Architecture where data is stored on the user's device as the primary source of truth
- Debt graph: Network visualization of financial obligations between group members
- Settlement optimizer: Algorithm that reduces N debts to minimum required transactions
- Health score: Proprietary 0–100 metric measuring a group's financial relationship quality
- PWA: Progressive Web App — installable web app with offline support

9.2 References

- Splitwise — splitwise.com (primary competitive reference)
- Linear — linear.app (design aesthetic reference)
- Notion — notion.so (UX simplicity reference)
- Minimalist → HUL acquisition (transparency brand benchmark)
- Google Gemini Flash API — ai.google.dev
- Dexie.js — dexie.org (local IndexedDB reference)

9.3 Document History

Version	Date	Author	Changes
v1.0	April 2026	Mann	Initial PRD — MVP scope, tech stack, full feature spec

PACT — Built for friends who trust each other.